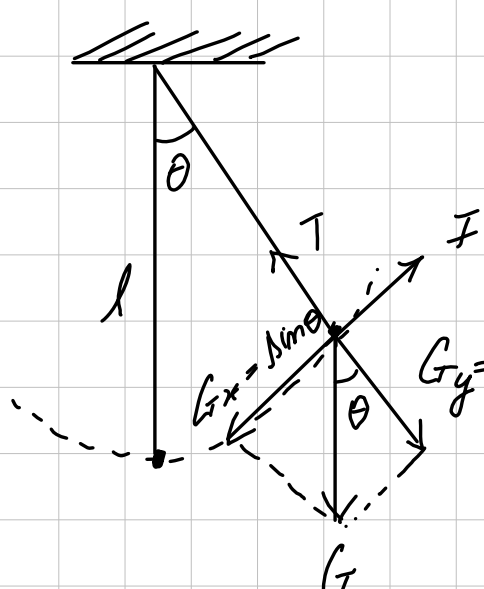


Pendulul

$$\vec{F} = \vec{G}_x$$

$$m \cdot a = -mg \sin \theta$$

$$a + g \sin \theta = 0$$

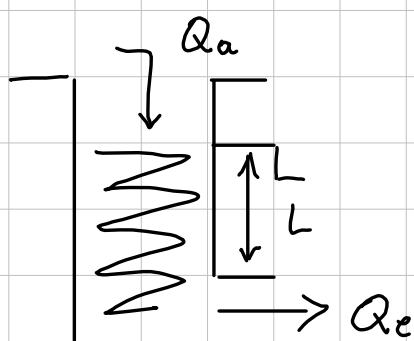
$$l \theta = x$$

$$a = \frac{d^2 x(t)}{dt^2}$$

$$\left. \begin{array}{l} \frac{d^2 \theta(t)}{dt^2} + g \sin \theta(t) = 0 \end{array} \right\}$$

$$\frac{d^2 \theta(t)}{dt^2} + g \theta(t) = 0$$

Det. nivelului unui lichid dintr-un rezervor



Q_a - det. intrare

Q_e - det. ieșire

L - max. l. rez(?)

L_0 -

A - sec. rezervă

reg. stat $Q_a = Q_e$

reg. dinamic $Q_a(t) - Q_e(t) = A \frac{dL(t)}{dt} \quad | \quad \alpha$

$$G(s) = \frac{L(s)}{Q_a(s)} \quad Q_a(s) = A s L(s) \Rightarrow G(s) = \frac{1}{A s}$$

$$b) Q_a(t) - Q_e(t) = A \frac{dL(t)}{dt}$$

$$Q_e(t) = \alpha \sqrt{2gL}$$

$$Q_a(t) - \alpha \sqrt{2gL(t)} = A \frac{dL(t)}{dt}$$

$$Q_e(t) = \underbrace{Q_e(t_0)}_{=0} + \underbrace{\frac{\partial Q_e(t)}{\partial L(t)}}_0 \bigg|_{L_0} (L(t) - L_0)$$

$$Q_a(t) - bL(t) = A \frac{dL(t)}{dt} \quad | \quad \alpha$$

$$Q_a(s) - bL(s) = A s L(s)$$

$$G(s) = \frac{L(s)}{Q_a(s)} = \frac{1}{A s + b}$$

$$\dot{x}_1 = x_1 + u = f_1(x_1, x_2)$$

$$\dot{x}_2 = -x_1 + x_1^2 = f_2(x_1, x_2)$$

$$y = x_2$$

$$\dot{\bar{x}} = A \bar{x} + B \bar{u}$$

$$(0, 0)$$

$$\bar{y} = C \bar{x} + D \bar{u}$$

$$(1, 0)$$

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{(x_1 s, x_2 s)}$$

$$\begin{bmatrix} \frac{d(x_1 - x_{1s})}{dt} \\ \frac{d(x_2 - x_{2s})}{dt} \end{bmatrix} = A \begin{bmatrix} x_1 - x_{1s} \\ x_2 - x_{2s} \end{bmatrix} + B u$$

$$y = Cx \quad \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = A \begin{bmatrix} x_1 - x_{1s} \\ x_2 - x_{2s} \end{bmatrix} + B u$$

$$y = Cx$$

$$\text{I. } \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1+2x & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = x_2$$

$$\dot{x}_1 = x_2 + u$$

$$\dot{x}_2 = -x_1$$

$$y = x_2$$

$$\text{II. } \dot{x}_1 = x_2 + u$$

$$\ddot{x}_2 = x_1$$

$$y = x_2$$